

MA3236 NONLINEAR PROGRAMMING

Semester 1, AY 2026/2027

Tutorial 1

Review: Lectures 1–2

- **Model and notation**

$$\min_{\mathbf{x} \in S} f(\mathbf{x}) \quad \text{or} \quad \max_{\mathbf{x} \in S} f(\mathbf{x}), \quad \max_S f = -\min_S(-f) \quad (\text{same optimizers}).$$

- $\mathbf{x}$: decision variable; f : objective function; S : feasible set.
- $\mathbf{x} \in S$: feasible; $\mathbf{x} \notin S$: infeasible.

- **Well-posedness for minimization**

$$S \neq \emptyset, \quad \inf_{\mathbf{x} \in S} f(\mathbf{x}) > -\infty, \quad \exists \mathbf{x}^* \in S : f(\mathbf{x}^*) = \inf_{\mathbf{x} \in S} f(\mathbf{x}).$$

- For maximization, replace inf and “bounded below” by sup and “bounded above”.
- S unbounded $\nRightarrow$ the objective is unbounded.

- **Graphical method in $\mathbf{R}^2$**

$$L_c := \{\mathbf{x} \in \mathbf{R}^2 : f(\mathbf{x}) = c\} \quad (\text{level curve}).$$

- Sketch S and representative L_c .
- Minimize: move toward smaller c ; maximize: toward larger c .
- Locate contact points, then check every relevant boundary piece.

- **Local versus global minimizer ($\mathbf{x}^* \in S$)**

$$\begin{aligned} \text{local: } & \exists \varepsilon > 0 : f(\mathbf{x}^*) \leq f(\mathbf{x}) \quad \forall \mathbf{x} \in S \cap B(\mathbf{x}^*, \varepsilon), \\ \text{global: } & f(\mathbf{x}^*) \leq f(\mathbf{x}) \quad \forall \mathbf{x} \in S. \end{aligned}$$

- global $\implies$ local; the converse need not hold.
- Strict: replace $\leq$ by $<$ for every other relevant point.

- **Closedness, compactness, and Weierstrass**

$$\begin{aligned} g, h \text{ continuous} & \implies \{\mathbf{x} : g(\mathbf{x}) = 0\}, \{\mathbf{x} : h(\mathbf{x}) \leq 0\} \text{ are closed,} \\ S \text{ compact} & \iff S \text{ closed and bounded} \quad (S \subseteq \mathbf{R}^n), \\ S \neq \emptyset, S \text{ compact, } f \text{ continuous} & \implies f \text{ attains both } \min_S f \text{ and } \max_S f. \end{aligned}$$

- **Coercivity and existence**

$$f \text{ coercive} \iff \|\mathbf{x}\| \rightarrow \infty \implies f(\mathbf{x}) \rightarrow +\infty.$$

$$S \neq \emptyset, S \text{ closed, } f \text{ continuous and coercive} \implies f \text{ attains } \min_S f.$$

- To disprove coercivity: find $\|\mathbf{x}^k\| \rightarrow \infty$ while $f(\mathbf{x}^k)$ does not tend to $+\infty$.
- Weierstrass and coercivity give sufficient, not necessary, conditions.

1. Consider the following constrained nonlinear programming problem.

$$\begin{array}{ll}\max & \frac{(x_1)^2}{4} + (x_2 - 1)^2 \\ \text{s.t.} & -1 \leq x_1 + x_2 \leq 1, \\ & -1 \leq x_1 - x_2 \leq 1.\end{array}$$

- (a) Sketch the feasible region S .
- (b) Solve the NLP graphically.

2. Consider the following nonlinear programming problem.

$$\begin{array}{ll}\text{maximize} & f(\mathbf{x}) := \frac{x_1^2}{4} + x_2^2 \\ \text{subject to} & x_1 + x_2 \leq 2 \\ & x_1 \geq 0, x_2 \geq 0\end{array}$$

- (a) Sketch the feasible set S .
- (b) Sketch the level curves correspond to $f(\mathbf{x}) = 1$ and $f(\mathbf{x}) = 9/4$. Hence determine the ‘direction’ where $f(\mathbf{x})$ increases.
- (c) Solve the NLP.

3. Consider the following constrained nonlinear programming problem.

$$\begin{array}{ll}\text{maximize} & x_1 + x_2 \\ \text{subject to} & -x_1^2 + x_2 \leq 0, \\ & -2 \leq x_1 \leq 2.\end{array}$$

- (a) Sketch the feasible region S .
- (b) Identify all local maximizers graphically.
- (c) Is there a global maximizer? If yes, determine all global maximizers.
- (d) Is it true that a local maximizer is a global maximizer? Justify your answer.

4. For each of the following NLPs, determine whether there is an optimal solution. Justify your answers. You need not find the optimal solution explicitly.

(a)

$$\begin{aligned} \min \quad & f(\mathbf{x}) := \frac{x_1^4 + x_1^3 x_2 + x_2^4 + 3}{x_1^4 + x_1 x_2^3} \\ \text{s.t.} \quad & (x_1 - 2)^2 + x_2^2 \leq 1 \\ & x_2 \geq 0 \end{aligned} \quad .$$

(b)

$$\begin{aligned} \min \quad & g(\mathbf{x}) := x_1^2 x_2 \\ \text{s.t.} \quad & x_1^2 - x_2^2 \leq 4 \\ & x_2 \geq 0 \end{aligned} \quad .$$

(c)

$$\begin{aligned} \min \quad & h(\mathbf{x}) := x_1 - x_2 \\ \text{s.t.} \quad & x_1^2 - x_2^2 = 4 \\ & x_2 \geq 0 \end{aligned} \quad .$$

(d)

$$\begin{aligned} \min \quad & q(\mathbf{x}) := x_1^4 + x_2^4 - x_1^2 x_2^2 - 2x_1^3 + x_2 \\ \text{s.t.} \quad & x_1 + x_2 \geq 1 \end{aligned} \quad .$$